

Jonesh Shrestha

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EDUCATION

DePaul University — *MS in Computer Science (AI), GPA: 3.84* Chicago, IL | Sep 2024 - Jul 2026
Relevant coursework: Artificial Intelligence I and II, Data Mining, Programming ML Applications, Distributed Systems, Applied Deep Learning

Kathmandu University — *BS in Computer Science* Nepal | Feb 2016 - Jun 2020

SKILLS

Languages: Python, SQL, R, Java, JavaScript
ML/AI: TensorFlow, scikit-learn, PyTorch, Pandas, NumPy, CNNs, Transformers, Computer Vision, NLP
Cloud & Big Data: AWS (EMR, S3, SageMaker), Apache Spark, Hadoop
Tools & Platforms: Git, Docker, Jupyter, Oracle Data Integrator (ODI), ETL Pipelines

PROFESSIONAL EXPERIENCE

DePaul University — *AI Research Assistant* Chicago, IL | Jul 2025 - Present

- Built and optimize deep learning pipelines for text detection, recognition, and role classification on scientific charts (CHART-Info 2024), turning static figures into structured, machine-readable data for downstream analytics.
- Evaluated and fine-tune OCR and rotated-object models (PaddleOCR, MMRotate with CNN/Transformer backbones), running controlled experiments and error analysis to improve detection quality and reduce missed/extra text regions in a shared research codebase.

Cotiviti Nepal Pvt. Ltd. — *Software Engineer* Nepal | Jul 2021 - Jul 2023

- Built ETL pipelines in SQL, PL/SQL, and Oracle Data Integrator to process US healthcare PHI for 15+ payer clients, standardizing vendor layouts and adding automated validation checks to ensure data quality during onboarding.
- Shipped production data workflows with cross-functional teams (developers, QA, PMs) and mentored 10+ engineers on ODI workflows, speeding up client integrations and change requests.

Kathmandu University — *Graduate Research Assistant* Nepal | Mar 2021 - Apr 2022

- Developed Nepali image captioning models using CNN and Transformer architectures, tuning attention and hyperparameters to improve BLEU and METEOR scores on a low-resource Nepali image-text dataset.

PROJECTS

US Healthcare Fraud, Waste & Abuse Detection — *End-to-end ML Pipeline* Oct 2025 - Nov 2025

- Developed a fraud detection pipeline on 558K Medicare claims (5.4K providers), aggregating data into 47 provider-level features and applying KMeans, Isolation Forest, Random Forest, and XGBoost.
- Tuned a Random Forest model to 95.46% ROC-AUC and 82.18% recall on fraudulent providers to prioritize high-risk providers for audit.

Developer Programming Language Prediction — *Production-ready ML Pipeline* Jun 2025 - Jul 2025

- Built a pipeline to predict developers' primary programming language from survey data, using grid-searched Decision Tree, Random Forest, and Logistic Regression models with k-fold CV, achieving 77.5% test accuracy.

Evolution Simulator — *Genetic Algorithm Simulation* Nov 2024 - May 2025

- Built a simulation where bipedal creatures evolve to walk using genetic algorithms. Featured on aijs.rocks.

ACHIEVEMENTS

- Published Paper: "Evaluating the Application of SOLID Principles in Modern AI Framework Architectures" arXiv (2025)
- Associate Member, Sigma Xi, The Scientific Research Honor Society (2025)